# Are we Targeting the Wrong Customers?

## Propensity vs Incrementality in Marketing Measurement

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## Background and Motivation

I have spent over a decade advising companies on marketing strategy, transformation, and digital performance marketing. One persistent tension in performance marketing is that **what is easiest to measure and optimize is not necessarily what the advertiser ultimately cares about**.

Digital advertising platforms generate enormous amounts of behavioral data and can predict conversion probability at remarkable scale. As a result, campaign optimization commonly focuses on finding consumers most likely to complete an action and attributing conversions to advertising exposures. But this creates a fundamental measurement and targeting problem: **A consumer who converts after seeing an ad is not necessarily a consumer who converted because of the ad.**

A high-propensity customer may have purchased anyway. Conversely, advertising may have its greatest incremental impact on someone with a lower baseline probability of converting. Optimizing for conversion propensity can therefore allocate advertising dollars toward customers who are easy to predict rather than customers whose behavior can actually be changed.

This distinction is increasingly relevant as marketers face pressure to demonstrate **incremental return**, rather than platform-attributed conversions or ROAS. IAB's recent measurement guidance similarly distinguishes incrementality, outcomes caused by marketing relative to what would otherwise have happened, from attribution-based measures of observed performance [1,2].

This project asks **whether better prediction of conversion necessarily leads to better marketing decisions**.

## Data

The project will use the publicly available **Criteo Uplift Prediction Dataset**, created from randomized digital-advertising incrementality experiments:

**Dataset:** https://ailab.criteo.com/criteo-uplift-prediction-dataset/

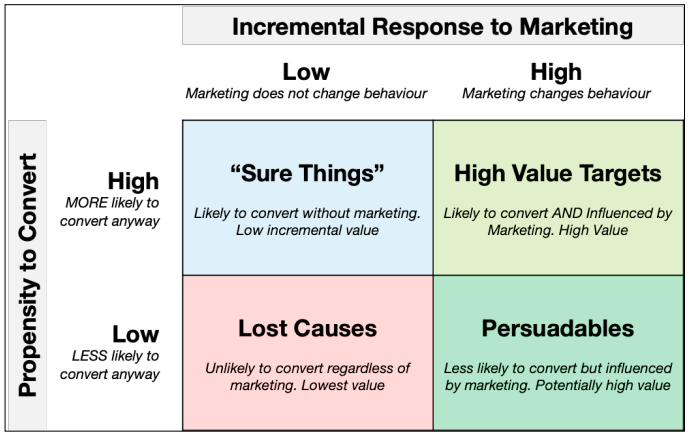
The dataset contains approximately **14 million observations** and 12 anonymized customer/context features (f0–f11). Crucially, it includes randomized advertising treatment/control assignment, actual advertising exposure, website visits, and conversions.

My preliminary inspection confirms approximately **11.9 million treatment observations, 2.1 million controls, 657,000 visits, and 41,000 conversions,** providing sufficient scale to study a rare conversion outcome.

## Problem

The primary research question will be: **Are the customers most likely to convert also the customers whom advertising is most likely to influence?**

Conceptually customers may fal into these 4 groups (see fig 1 below):




A conventional propensity strategy prioritizes the top row. An incrementality-oriented strategy should prioritize the right-hand column.

The project will investigate how closely those strategies align and, ultimately, ask: If an advertiser can reach only a limited share of consumers, does targeting predicted converters or predicted incremental converters produce greater incremental impact?

The project will compare **baseline conversion propensity** with **incremental advertising response** and investigate whether conventional propensity-based targeting identifies the customers marketers should prioritize.

A secondary business question is whether, under a constrained advertising budget, targeting consumers based on estimated incrementality could generate more incremental conversions than targeting those with the highest conversion propensity.

## Scope and Methods

First, I propose estimating **baseline conversion propensity**, ideally using the randomized control population to model the probability of conversion without advertising. **Logistic regression** will provide an interpretable baseline, followed by course methods such as **regularized logistic regression, decision trees, random forests, and boosting**. Given the approximately 0.3% conversion rate, evaluation will emphasize metrics appropriate for severe class imbalance rather than accuracy alone.

Second, randomized treatment/control outcomes will be used to examine whether advertising lift varies across baseline-propensity groups. If feasible, separate treatment and control response models can provide a simple estimate of individual uplift:

predicted conversion with advertising – predicted conversion without advertising.

Finally, the project can compare simulated targeting policies under the same audience constraint: **random targeting, conversion-propensity targeting, and incrementality-based targeting**. The business outcome is therefore not simply which algorithm achieves the highest predictive accuracy, but which targeting objective generates greater incremental response from a fixed marketing opportunity.

## Concerns and Limitations

The largest limitation is interpretability. Criteo deliberately anonymizes the 12 customer/context features, preventing conclusions such as *which demographic or behavioral characteristics make a consumer persuadable*. This limits customer-level storytelling, although it also resembles the opacity advertisers increasingly encounter when platform algorithms optimize audiences using signals advertisers cannot directly inspect.

The outcome is extremely imbalanced, and the 3GB dataset creates computational considerations that may require sampling or chunk-based processing. Estimating heterogeneous treatment effects is also more difficult than conventional prediction, so sophisticated causal-ML methods will be treated as extensions rather than required for project success.

Finally, **predictability, persuadability, and ethical desirability are different questions**. Even if incrementality-based targeting improves marketing efficiency, determining whose behavior can most easily be changed raises broader questions about responsible personalization and algorithmic influence.

## References

[1] Interactive Advertising Bureau (2026). State of Data 2026: The AI-Powered Measurement Transformation.

[2] IAB & IAB Europe (2025). Guidelines for Incremental Measurement in Commerce Media.